Why There Are Chemical Structures At All: Bonds, Valence, and Molecular Geometry as Boundary-Thickness Minimisation Among Individuated Atoms

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Abstract

We ask why there are chemical structures at all—why atoms compose into molecules of definite stoichiometry and shape rather than remaining separate or merging without limit—and answer it from a single hypothesis already known to force atomic structure: that an atom is a *bounded, partitionable system whose individuation from the rest of a non-completable whole carries a strictly positive cost*, the contact floor $\beta > 0$. We take as given the partition structure of the isolated atom—the coordinates (n, ℓ, m, s), the shell capacity $C(n) = 2n^2$, and the resulting electron configurations—and add nothing to it. From the floor alone we derive a chain of existence results. **(C1)** Every individuated atom carries a boundary of positive thickness; a closed-shell atom realises the local minimum thickness (a noble gas), while an open-shell atom carries excess thickness proportional to its number of unfilled partition states—its *vacancy*, the quantitative form of "partition malformation." **(C2)** Two atoms bond *if and only if* their joint partition has lower total boundary thickness than the separated atoms; bonding is not attraction added to atoms but the same boundary-minimisation that individuated them, now run between them. **(C3)** The number of bonds an atom forms—its *valence*—is its vacancy count, and the stable molecule is the one that drives every constituent's vacancy to the shell-closure minimum: the octet and duet rules are the boundary-minimum condition, not empirical regularities. **(C4)** A bond has a definite length: the joint boundary thickness, convex in internuclear separation, has a unique minimiser at finite distance, neither zero (the floor forbids coincidence) nor infinity (separation forgoes the reduction). **(C5)** A molecule has a definite shape: because individuation in three dimensions must close around each centre from all angular directions, the bonds of a multivalent atom adopt the maximally separated angular configuration, yielding the observed geometries. **(C6)** Reaction is the propagation of boundary toward its joint minimum, irreversible by the monotone committed-boundary count; affinity is available thickness reduction and equilibrium is the floor-limited halt. We illustrate the chain on H_2, the noble gases, NaCl, HCl, LiF, H_2O, NH_3, CH_4, CO_2 and benzene, deriving in each case why the structure exists, its stoichiometry, and its geometry, with no parameter beyond the shell structure of the isolated atoms. The account explains, as one fact, why noble gases are inert, why halogens and alkali metals are reactive, why valences are what they are, why water bends and carbon is tetrahedral, and—most basically—why a world of bounded atoms is compelled to be a world of molecules.

Keywords: chemical bond; valence; molecular geometry; contact floor; boundary thickness; partition malformation; octet rule; individuation by negation; bounded phase space; non-completable whole.

Contents

1 Introduction

1.1 The question

Chemistry begins with a fact so familiar it is rarely posed as a question: matter is not a uniform soup of atoms, nor an unbounded agglomeration, but a definite architecture of *molecules*—atoms joined in fixed proportions and fixed shapes. Hydrogen appears as H_2, never H_3; oxygen pulls two hydrogens and no more; carbon arranges four neighbours at the corners of a tetrahedron; neon refuses to combine with anything. Why? The conventional answer assembles several independent ingredients—a Schrödinger equation for the atom, a variational treatment of the molecule, an empirical octet heuristic, a valence-shell-electron-pair-repulsion rule for shape, a thermodynamic criterion for whether a reaction proceeds. Each works; none explains why the others should hold, and none answers the prior question of why there should be discrete structures at all rather than none.

This paper offers a single answer to that prior question. It rests on one hypothesis, and the hypothesis is not new: it is the same one that, in companion developments, forces the isolated atom to have the structure it has. The hypothesis is that to be an individuated thing—an atom told apart from the rest of the world—costs a strictly positive minimum, and that this cost is a measurable boundary. We show that chemical structure is what a world of such atoms is compelled to form, because composing into molecules is how bounded atoms *lower* that cost. Bonds, valence, length, shape, and reactivity are not five facts; they are five readings of one minimisation.

1.2 What is assumed, and what is derived

We assume the results of the partition-geometric account of the isolated atom and treat them as a black box: a persistent bounded system admits partition coordinates (n, ℓ, m, s) with shell capacity $C(n) = 2n^2$, and filling these under exclusion and energy ordering yields the electron configurations and shell-closure pattern of the elements (Griffiths and Schroeter, 2018; Madelung, 1936; Pauli, 1925). We add no new postulate about atoms. The single structural fact we carry forward and use as the engine is the *contact floor*: individuating any part of a bounded whole deposits boundary content bounded below by a positive constant $\beta > 0$, and this floor is itself forced by the non-completability of the whole against which a thing is individuated. Everything about *molecules* below is derived from the floor together with the atom's shell structure—nothing else.

1.3 The thesis in one line

A molecule exists because two or more bounded atoms, each carrying a boundary it cannot abolish, can share boundary and thereby hold a *lower total thickness together than apart.* Chemical structure is boundary-thickness minimisation among individuated atoms; its stoichiometry is the vacancy arithmetic of shell closure; its geometry is the angular form that minimisation must take in three dimensions; its dynamics is the monotone propagation of boundary toward the joint minimum. There are chemical structures at all because, for bounded atoms, being joined is cheaper than being separate—and the floor guarantees the joining never collapses to a point.

1.4 Relation to existing accounts

The framework is not a rival calculation of bond energies; standard quantum chemistry computes those to high accuracy and we do not improve on it (Levine, 2014; Szabo and Ostlund, 1996). It is an account of *why* the qualitative architecture is forced—why there are integer valences, why closed shells are inert, why shapes are definite—placing those facts on a single footing rather than treating each as a separate rule. Where the octet rule (Langmuir, 1919; Lewis, 1916) is normally an empirical summary and VSEPR (Gillespie and Nyholm, 1957) a mnemonic for shape, here both are corollaries of one minimisation. The relationship to standard theory is that of a variational principle to its solutions: we state the quantity that chemistry minimises and read off the qualitative consequences; the quantitative solution remains the province of the electronic-structure methods.

2 The inherited atom and the contact floor

We fix the objects carried over from the atomic theory and state the one engine we add.

2.1 The atom as a bounded partition

Definition 2.1 (Atomic partition)**.** An *atom* of atomic number Z is a bounded partitionable system whose accessible states are labelled by partition coordinates (n, ℓ, m, s) with $n \in \mathbb{N}^+$, $0 \leq \ell \leq n-1$, $-\ell \leq m \leq \ell$, $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$, and whose shell at depth n holds

$$C(n) = \sum_{\ell=0}^{n-1} (2\ell + 1) \cdot 2 = 2n^2$$

states. Filling Z states under exclusion and energy ordering $E \propto n + \alpha\ell$ gives the atom's configuration; the outermost partially or fully occupied shell is its *valence shell*, of capacity $\Cap_v$ and occupancy q_v.

Definition 2.2 (Shell closure and vacancy)**.** The valence shell is *closed* if $q_v = \mathbb{m}_v$ (all valence states occupied) and *open* otherwise. The *vacancy* of an atom is

$$\nu \;:=\; \mathbb{m}_v - q_v \;\in\; \{0, 1, \ldots, \mathbb{m}_v\},$$

the number of unfilled valence states; its *saturation deficit* is ν, and a closed shell has $\nu = 0$.

For the light elements the relevant valence capacities are $\mathbb{m}_v = 2$ (the $n = 1$ shell, "duet") and $\mathbb{m}_v = 8$ (the $2s\,2p$ and $3s\,3p$ valence octets); we use these throughout the worked cases and note that the arithmetic is identical for any $\mathbb{m}_v$.

2.2 The contact floor as the engine

Principle 2.3 (Contact floor)**.** Individuating a bounded part A from the rest of a non-completable whole M deposits boundary content of strictly positive measure. There is a constant $\beta > 0$, the *contact floor*, such that the boundary thickness $B(A)$ of any individuated part satisfies $B(A) \geq \beta > 0$. The floor is not a posit but the trace the inexhaustible whole leaves on each finite part: a completed comparison of A against $\mathsf{M} \setminus A$ would exhaust M, which non-completability forbids, so an irreducible residual remains.

Remark 2.4 (What "boundary thickness" means here)*.* For an atom, the boundary is the partition surface separating its occupied valence structure from the surrounding medium of other atoms. Its *thickness* B is the content of that separator: the cost of maintaining the atom as a distinct individual against its environment. We do not need a closed-form B; the entire derivation uses only three of its properties, each inherited or elementary: (i) $B \geq \beta > 0$ always (Principle 2.3); (ii) B is larger for an open shell than for the closed shell of the same period, by an amount increasing in the vacancy ν (Lemma 2.5); and (iii) sharing a boundary between two atoms is subadditive (Lemma 3.2). All quantitative chemistry sits inside property (ii)'s constant of proportionality, which we do not compute; the *qualitative architecture* follows from the three properties alone.

Lemma 2.5 (Thickness increases with vacancy)**.** *Among atoms of a given valence capacity $\mathbb{m}_v$, the boundary thickness is a strictly increasing function of the vacancy: $B = B_0 + \kappa\,\phi(\nu)$ with $B_0 \geq \beta$ the closed-shell thickness, $\kappa > 0$, and ϕ strictly increasing with $\phi(0) = 0$.*

Proof. An unfilled valence state is a partition cell that the atom's boundary must enclose without occupying—a place where the separator between the atom and the medium is incompletely committed, i.e. a locally thicker, less-resolved boundary (an open separator in the sense that the cell borders both the atom's occupied structure and the medium). Each additional vacancy adds one such incompletely-committed cell to the boundary, so the separator content is monotone in ν. A closed shell ($\nu = 0$) has every valence cell committed and realises the minimum thickness B_0 of its period; hence $\phi(0) = 0$ and ϕ strictly increasing. Positivity of B_0 is Principle 2.3. □

Remark 2.6 (Malformation is positive vacancy)*.* An open-shell atom is, in the language of the atomic theory, a *partition malformation*: its categorical structure is incomplete, carrying excess boundary $\kappa\,\phi(\nu) > 0$ above the closed-shell minimum. Chlorine ($\nu = 1$ in an octet shell) is malformed by one vacancy; argon ($\nu = 0$) is not malformed at all. The thesis of the paper is that chemistry is the relaxation of these malformations by shared boundary, and that the relaxation cannot reach zero because of the floor.

3 C1–C2: why bonds exist

3.1 Shared boundary is subadditive

Definition 3.1 (Joint individuation and shared boundary)**.** Two atoms A, B brought into contact may be individuated *jointly*: the composite $A \oplus B$ is individuated from the medium by a single outer boundary, while A and B share an internal partition surface. The *shared boundary* is the portion of A's and B's separators that, in the joint individuation, faces the other atom rather than the medium.

Lemma 3.2 (Subadditivity of shared boundary)**.** *Let $B(A)$, $B(B)$ be the thicknesses of the separately individuated atoms and $B(A \oplus B)$ the total thickness of the jointly individuated pair. Then*

$$B(A \oplus B) \;\leq\; B(A) + B(B),$$

with strict inequality whenever A and B share a boundary of positive content. The reduction is

$$\Delta B(A, B) \;:=\; B(A) + B(B) - B(A \oplus B) \;\geq\; 0,$$

the shared content*.*

Proof. When A and B are individuated separately, each maintains a full separator against the medium; the cells where A faces B are counted in $B(A)$ and the cells where B faces A are counted in $B(B)$, so the interface is paid for twice. In the joint individuation the interface is a single shared surface: the cells between A and B are committed once, jointly, rather than twice, separately. The total separator content therefore drops by the content of the doubly-counted interface, which is $\Delta B(A, B) \geq 0$, strict when the interface has positive content. No cell is removed below the floor: the outer boundary of $A \oplus B$ against the medium still satisfies Principle 2.3. □

Remark 3.3 (Bonding is the same act as individuation)*.* Lemma 3.2 is the crux. A bond is not an attractive force added on top of two finished atoms; it is the recognition that two atoms can be individuated by *one* shared boundary more cheaply than by two separate ones. The same boundary-minimisation that made each atom a distinct thing, run *between* the atoms, is the bond. This is why no new postulate is needed: chemistry is the floor's own logic applied to more than one centre.

3.2 The bonding criterion

Theorem 3.4 (Bonds exist iff they lower total thickness)**.** *Two atoms A, B form a bond if and only if the jointly individuated pair has strictly lower total boundary thickness than the separated atoms:*

$$\boxed{A\text{–}B \text{ bonds} \quad\Longleftrightarrow\quad B(A \oplus B) < B(A) + B(B),}$$

equivalently iff the shared content $\Delta B(A, B) > 0$. Moreover the shared content available is bounded by the atoms' vacancies: an interface can commit at most $\min(\nu_A, \nu_B)$ valence cells jointly, so $\Delta B(A, B)$ increases with the vacancy each atom brings to the interface and is zero when either atom is closed-shell.

Proof. (Existence direction.) A configuration is realised when it lowers the boundary an individuation must maintain (the system takes the cheaper individuation; this is the boundary-minimisation principle that selects states, used already to fill the atom). If $B(A \oplus B) < B(A) + B(B)$ then the joint individuation is cheaper than the separate ones, so the joined configuration is the realised one: a bond. If $B(A \oplus B) \geq B(A) + B(B)$ there is no thickness reduction to be had and the separate individuation is at least as cheap, so no bond forms.

(Vacancy bound.) By Lemma 3.2 the shared content is the content of the interface committed jointly. An interface cell is a valence cell of A paired with a valence cell of B across the shared surface; a closed-shell atom has no unfilled valence cell to offer to the interface, so it can share none—$\Delta B = 0$ and no bond. An open atom offers up to ν cells, so the interface commits at most $\min(\nu_A, \nu_B)$ paired cells, and by Lemma 2.5 each committed pair removes excess thickness $\kappa[\phi(\nu) - \phi(\nu - 1)] > 0$ from both atoms. Hence ΔB grows with the shared vacancy and vanishes when either atom is closed. □

Corollary 3.5 (Noble gases are inert; open shells are reactive)**.** *A closed-shell atom ($\nu = 0$) has $\Delta B(A, B) = 0$ with every partner, so forms no bonds: it is inert. An open-shell atom ($\nu > 0$) has $\Delta B > 0$ with any partner that can pair its vacancies, so is reactive, the more so the larger its vacancy and the better matched the partner. The chemical inertness of He, Ne, Ar, Kr, Xe and the reactivity of H, the alkali metals, and the halogens are the two sides of Theorem 3.4.*

Proof. Immediate from the vacancy bound: $\nu = 0 \Rightarrow \Delta B = 0$ for all partners (inert); $\nu > 0 \Rightarrow \Delta B > 0$ for a vacancy- matching partner (reactive). The magnitude orders with vacancy and match. □

Example 3.6 (H_2 exists; He_2 does not)**.** Hydrogen has valence capacity $\mathbb{m}_v = 2$ (the $n = 1$ duet) and occupancy 1, so $\nu_{\mathrm{H}} = 1$. Two hydrogens each bring one vacancy to the interface; the shared surface commits one paired cell, driving each atom's valence occupancy to $2 = \mathbb{m}_v$: both reach closure through the shared boundary. $\Delta B > 0$, so H_2 forms, and no third hydrogen can bond because both vacancies are spent ($\nu = 0$ in the molecule). Helium has $\mathbb{m}_v = 2$, occupancy 2, $\nu_{\mathrm{He}} = 0$: no vacancy to share, $\Delta B = 0$, no He_2. The existence of H_2 and the non-existence of He_2 are one calculation.

4 C3: valence and the closure (octet/duet) rule

4.1 Valence is the vacancy count

Theorem 4.1 (Valence equals vacancy)**.** *The number of bonds an atom forms in a thickness-minimising molecule—its* valence*—equals the number of valence cells it must commit through shared boundaries to reach closure, which for an atom short of closure is its vacancy ν, and for an atom past half-closure may instead be the co-vacancy $\mathbb{m}_v - \nu$ when sharing electrons it already holds is cheaper than acquiring the full ν. In the light-element octet/duet regime the realised valence is*

$$\text{valence} = \min(\nu,\ \mathbb{m}_v - \nu) \textit{ for the sharing (covalent) interface,}$$

and the molecule is stable when every constituent reaches $\nu = 0$.

Proof. By Theorem 3.4 each bond commits one paired valence cell and removes the excess thickness of one vacancy from each partner. To reach the closed-shell minimum ($\nu = 0$, Lemma 2.5) an atom must have all its vacancies paired; the number of bonds needed is therefore the number of vacancies to be filled. An atom may reach closure either by acquiring its ν missing cells (sharing into its vacancies) or, equivalently for the count, by sharing out the $\mathbb{m}_v - \nu$ cells that would otherwise need separate individuation; the cheaper of the two sets the realised covalent valence, $\min(\nu, \mathbb{m}_v - \nu)$. The molecule as a whole is at its thickness minimum when no constituent carries residual vacancy, i.e. when every valence shell is closed—each constituent at $\nu = 0$. □

Corollary 4.2 (The octet and duet rules are the boundary minimum)**.** *For valence capacity $\mathbb{m}_v = 8$, a thickness-minimising molecule drives every main-group constituent to eight committed valence cells (the octet); for $\mathbb{m}_v = 2$, to two (the duet, e.g. hydrogen). These are not empirical rules but the statement that the molecular boundary is minimal exactly when every constituent shell is closed.*

Proof. Closure $\nu = 0$ is the per-atom thickness minimum (Lemma 2.5); the molecular total is minimised when each term is minimised and the shared interfaces are maximised (Theorem 3.4). For $\mathbb{m}_v = 8$ this is eight committed cells per atom; for $\mathbb{m}_v = 2$, two. Hence the octet and duet. □

Example 4.3 (Valences of the second period)**.** Carbon ($2s^2 2p^2$, $q_v = 4$, $\mathbb{m}_v = 8$): $\nu = 4 = \mathbb{m}_v - \nu$, valence 4. Nitrogen ($q_v = 5$): $\nu = 3$, valence 3. Oxygen ($q_v = 6$): $\nu = 2$, valence 2. Fluorine ($q_v = 7$): $\nu = 1$, valence 1. Neon ($q_v = 8$): $\nu = 0$, valence 0 (inert). These are the observed valences, read directly off the vacancy arithmetic with no further input.

4.2 Worked stoichiometries

Example 4.4 (NaCl, HCl, LiF: vacancy matching across the interface)**.** Sodium ([Ne]$3s^1$, octet valence shell, ν effectively the single $3s$ electron above a closed Ne core; sharing *out* that one cell closes its lower octet) and chlorine ([Ne]$3s^2 3p^5$, $\nu = 1$) each have one cell to settle. One shared interface closes both: Na reaches the Ne configuration, Cl reaches the Ar configuration, $\Delta B > 0$, stoichiometry 1:1—NaCl. Identically, HCl pairs hydrogen's one duet vacancy with chlorine's one octet vacancy (1:1), and LiF pairs lithium's one with fluorine's one (1:1). In each case the stoichiometry is forced by $\min(\nu_A, \nu_B)$ matching: one vacancy on each side, one bond, 1:1.

Example 4.5 (Why H_2O is 2:1 and not 1:1)**.** Oxygen has $\nu = 2$; hydrogen has $\nu = 1$. A single O–H interface closes hydrogen but leaves oxygen with one residual vacancy and thus non-minimal thickness. Minimisation requires oxygen's second vacancy also committed: a second hydrogen. Two O–H interfaces drive oxygen to $\nu = 0$ and both hydrogens to $\nu = 0$ simultaneously—H_2O. A third hydrogen finds no vacancy on a closed oxygen ($\Delta B = 0$), so H_3O does not form as a neutral molecule. The formula 2:1 is the vacancy ratio $\nu_{\mathrm{O}} : \nu_{\mathrm{H}} = 2 : 1$.

Example 4.6 (NH_3, CH_4, CO_2)**.** Nitrogen ($\nu = 3$) closes by three N–H interfaces: NH_3. Carbon ($\nu = 4$) closes by four C–H interfaces: CH_4. Carbon with oxygen: carbon must commit four cells, each oxygen two; two oxygens supply four cells total and each oxygen closes by committing both of its cells to the same carbon (a double interface, two paired cells per C–O), giving O=C=O, CO_2, with carbon at $\nu = 0$ and both oxygens at $\nu = 0$. The stoichiometry and the double interfaces are both fixed by closing every vacancy at minimum total thickness.

5 C4: why a bond has a definite length

Theorem 5.1 (Bond length is the thickness minimiser)**.** *For a bonding pair A, B, the total boundary thickness $B(A \oplus B; r)$ as a function of internuclear separation r is strictly convex with a unique minimiser $r^* \in (0, \infty)$. The bond length is r^*: neither* 0 *(coincidence) nor* ∞ *(separation).*

Proof. Two competing terms set the r-dependence of the joint thickness. As r decreases from infinity the shared interface grows, so the shared content $\Delta B(r)$ increases and the total thickness falls: a decreasing contribution. But the floor forbids the two atoms' boundaries from coinciding—driving $r \to 0$ would require committing a separator of content below β between the two centres (two distinct individuated nuclei cannot share a sub-floor boundary, Principle 2.3), so the cost of compression diverges as $r \to 0$: an increasing contribution that blows up. The total is a sum of a term decreasing in the interface (saturating as the shells close) and a term diverging as $r \to 0$; such a sum is convex on $(0, \infty)$ with $B \to +\infty$ as $r \to 0$ and $B \to B(A) + B(B)$ as $r \to \infty$, hence attains a unique interior minimum at some $r^* \in (0, \infty)$. That r^* is the equilibrium bond length. □

Remark 5.2 (The floor is the repulsive wall). Theorem 5.1 identifies the origin of the short-range repulsion that every bond model needs: it is the contact floor. Two nuclei cannot be individuated at zero separation because that would be a zero-cost boundary, which the floor forbids; the divergence of thickness as $r \to 0$ is the floor asserting itself. The bonded state is the standoff between boundary-sharing (which pulls the atoms together) and the floor (which forbids coincidence)—the same two facts, attraction-by- sharing and a hard floor, that a Lennard-Jones or Morse curve encodes phenomenologically, here derived rather than fitted.

6 C5: why a molecule has a definite shape

6.1 Why three dimensions: no axis is privileged

Before the angular argument, we must say why the surrounding space in which a centre is individuated is three-dimensional, because the geometry of a molecule is downstream of that fact.

The reason is the same selector-free condition that drove individuation throughout: an atom is fixed by negation against everything it is not, with *no external chooser*. We show three dimensions is the minimum at which this condition can be met spatially.

Principle 6.1 (Three dimensions from the absence of a privileged axis)**.** A space of orthogonal coordinate axes individuates a centre selector-free only if no axis is privileged—if the assignment of which direction plays which coordinate role is fixed by the system internally, not imposed from outside. The minimum dimension in which this holds is $d = 3$:

(i) In $d = 2$, two perpendicular axes carry fixed, distinct roles ("horizontal" and "vertical"). Exchanging the two roles is the map $(x, y) \mapsto (y, x)$, a reflection: it has determinant -1 and is *not* a rotation of the plane (not an element of $SO(2)$). There is no continuous proper motion within the plane that turns one axis into the other, so which axis is which must be *stipulated from outside*—an external selector. A two-dimensional individuation therefore smuggles in the very chooser the framework forbids.

(ii) In $d = 3$, the addition of a third axis makes the exchange of any two axes a *proper rotation*: the 180° rotation about the axis bisecting them, e.g. $(x, y, z) \mapsto (y, x, -z)$, lies in $SO(3)$ (determinant $+1$) and is realised by a continuous internal motion. The coordinate roles become mutually interchangeable by the system itself; no axis is privileged, and no orientation need be imposed from outside.

(iii) Hence $d = 3$ is the least dimension in which a centre can be individuated by negation with no external selector. A one- or two-dimensional world cannot host selector-free individuation; a three-dimensional one can, and no further dimension is required to remove a privileged axis.

Remark 6.2 (Why this is the spatial form of "no selector")*.* The founding move of the whole account is that a thing is individuated without a privileged point or chooser (individuation by negation). Principle 6.1 is exactly that move applied to the coordinate frame: in $d < 3$ the frame itself carries a privileged direction that must be chosen externally, contradicting selector-freedom; $d = 3$ is the first dimension in which the frame's directions are exchangeable by an internal proper rotation, so the frame imposes no choice. Three-dimensionality is therefore not an extra postulate of the chemical theory—it is what selector-free individuation requires of the space, and the bond geometries below are its consequence.

6.2 Negation closes around a centre with no preferred direction

Principle 6.3 (Angular closure of individuation)**.** Because no axis is privileged in three dimensions (Principle 6.1), individuating a centre from its surroundings by negation closes around the centre from *all* angular directions equally: a thing is fixed by what it is not, "everything it is not" surrounds it on a two-sphere, and no direction on that sphere is cheaper or more privileged than any other. A bonded neighbour is a committed portion of a centre's boundary occupying a solid-angle region; two neighbours occupying overlapping solid angles share boundary that is then doubly committed, raising thickness. The absence of a privileged direction is what forbids the bonds from preferring one axis and forces them to distribute over the sphere.

Theorem 6.4 (Bonds adopt the maximally separated angular configuration)**.** *The k bonded interfaces of a k-valent centre minimise total boundary thickness when their solid-angle regions are as mutually separated as possible on the surrounding two-sphere; i.e. the bond directions are the configuration of k points on the sphere of maximal mutual angular separation. Consequently:*

$$k = 2: \ \textit{linear}\ (180^\circ); \quad k = 3: \ \textit{trigonal planar}\ (120^\circ); \quad k = 4: \ \textit{tetrahedral}\ (109.5^\circ).$$

Proof. By Principle 6.3 each bonded interface occupies a solid-angle region of the centre's boundary, and—because no direction on the two-sphere is privileged (Principle 6.1)—no interface can lower thickness by preferring a special axis; the only thickness-relevant quantity is the mutual overlap between regions. Two interfaces whose regions overlap share boundary content that is committed for both, adding excess thickness proportional to the overlap (the doubly-committed cells of Lemma 3.2, now between two neighbours of the same centre). Total thickness is therefore minimised by minimising pairwise solid-angle overlap, and with no axis privileged the minimum is the *symmetric* one: the k interface directions placed at maximal mutual angular separation on the two-sphere. The maximal-separation configurations of k points on the sphere are, for $k = 2, 3, 4$, the antipodal pair (180°), the equatorial triangle (120°), and the regular tetrahedron ($\cos^{-1}(-\frac{1}{3}) = 109.47°$). These are the bond geometries. □

Example 6.5 (CH_4, NH_3, H_2O: the same tetrahedral closure)**.** Carbon in CH_4 has four committed interfaces and no residual vacancy; maximal separation of four interfaces is the tetrahedron, 109.5°—the observed methane geometry. Nitrogen in NH_3 has three bonded interfaces and one committed-but-unshared region (its residual valence pair); four regions in all still seek tetrahedral separation, so the three N–H bonds sit at the base of a tetrahedron, compressed slightly below 109.5° by the larger unshared region—the observed 107°. Oxygen in H_2O has two bonded interfaces and two unshared regions; four regions, tetrahedral base, two bonds compressed further to the observed $\approx 104.5°$. The bent shape of water, the pyramidal shape of ammonia, and the tetrahedral shape of methane are one closure condition with different numbers of bonded versus unshared regions.

Example 6.6 (CO_2 is linear, benzene is hexagonal)**.** Carbon in CO_2 has two bonded interfaces (each a double C=O); two regions separate maximally to 180°—linear O=C=O. In benzene each carbon commits three interfaces (two ring C–C and one C–H) at 120° trigonal separation; six such trigonal centres close into a planar regular hexagon, the only ring that lets every carbon hold 120° while closing the ring—the observed benzene geometry. Shape in every case is the angular form boundary-minimisation must take in three dimensions.

7 C6: reaction, affinity, equilibrium

Principle 7.1 (Reaction is boundary propagation toward the joint minimum)**.** A chemical reaction is the propagation of boundary among a set of atoms from a higher-total-thickness configuration to a lower one: the system re-individuates its parts so as to lower total boundary thickness. A reaction proceeds if and only if a re-individuation of strictly lower total thickness is available; its driving magnitude—the *affinity*—is the available thickness reduction.

Theorem 7.2 (Reaction direction, irreversibility, and equilibrium)**.** *Let a set of atoms admit configurations $\mathcal{C}$ with total thickness $B(\mathcal{C})$. Then:*

(i) ***Direction.*** *A transition $\mathcal{C} \to \mathcal{C}'$ occurs only if $B(\mathcal{C}') < B(\mathcal{C})$; the affinity is $B(\mathcal{C}) - B(\mathcal{C}') > 0$. Adding a reagent (e.g. fluorine to a system) drives a reaction precisely when it opens a lower-thickness re-individuation, and not otherwise.*

(ii) ***Irreversibility.*** *The committed-boundary count is monotone non-decreasing: each re-individuation commits new shared cells, and un-committing them is itself a further committing step. No reaction returns the system to a prior individuation state; resemblance of configuration is not identity of state.*

(iii) ***Equilibrium.*** *The reaction halts not at zero thickness but at the floor-limited minimum: it proceeds while a strictly lower attainable configuration exists and stops when the marginal thickness reduction available no longer exceeds the floor cost of the next committing step. Equilibrium is the floor-limited halt, never coincidence of parts.*

Proof. (i) is Principle 7.1 applied to the configuration set: the realised transition is the one lowering total thickness, with affinity the reduction. The fluorine clause is the special case where the reagent's vacancies open shared interfaces unavailable before its addition: if they lower the total, the reaction runs; if not, none does. (ii): committing a shared interface adds cells to the committed-boundary record, which is a count that increases by at least one per committing step; reversing a commitment is a distinct committing step (re-individuation), so the count never decreases—the monotone-record / non-return property of individuation in a bounded whole. (iii): by the floor, thickness cannot be driven below β per maintained boundary, and the cost of further compression diverges (the convex wall of Theorem 5.1); a finite system therefore halts where marginal reduction meets marginal floor cost—a strictly positive-thickness equilibrium, not the coincidence $B \to 0$, which the floor forbids. □

Remark 7.3 (One quantity under five names). Theorems 3.4, 4.1, 5.1, 6.4, and 7.2 are five readings of the single functional B: a bond is where B drops on joining, valence is how many drops are available, length is where the drop is maximal in r, shape is the angular arrangement that maximises the drop, and reaction is the propagation that realises the drop. There are not five chemical principles; there is one boundary-thickness minimisation, seen from five sides. This is the sense in which the account unifies what are normally separate rules.

8 Why there are chemical structures at all

We can now state the answer to the title question as a single conditional, parallel to the way structure-in-general was shown to follow from a non-completable whole.

Theorem 8.1 (Existence of chemical structure)**.** *Let a world consist of finitely many bounded atoms, each individuated from a non-completable whole and so carrying boundary thickness $B \geq \beta > 0$, and let some atoms be open-shell ($\nu > 0$). Then chemical structures necessarily exist: there are configurations of strictly lower total thickness than the all-separate configuration (any vacancy-matched pair, by Theorem 3.4), the world relaxes toward them (Theorem 7.2), and the relaxed configurations are molecules of definite stoichiometry (Theorem 4.1), length (Theorem 5.1), and shape (Theorem 6.4). Conversely, a world of only closed-shell atoms ($\nu = 0$ throughout) forms no structures—it is a noble-gas world, inert and structureless. Chemical structure exists if and only if there is uncommitted boundary (vacancy) to share.*

Proof. If some atom has $\nu > 0$, then by Theorem 3.4 it has a partner with $\Delta B > 0$, so a strictly lower-thickness joined configuration exists; by Theorem 7.2(i) the world transitions toward it, and by (iii) halts at a floor-limited minimum holding the shared interfaces—a molecule, whose count, length, and shape are fixed by Theorems 4.1, 5.1, 6.4. If every atom is closed-shell, then $\Delta B = 0$ for all pairs (Corollary 3.5); the all-separate configuration is already minimal and no structure forms. Hence chemical structure $\iff$ positive shareable vacancy. □

Remark 8.2 (The shadow of the floor, one level up). The companion account showed that structure-in-general is the shadow a non-completable whole casts on its finite parts: because individuation costs a positive floor, parts are forced, bounded, and never reach a point identity. The present result is that same shadow cast one level up. Atoms are the individuated parts; their floors are positive; and *because* the floor cannot be abolished but *can* be shared, bounded atoms are driven to compose. A molecule is two or more atoms holding a single boundary where two would otherwise be paid for. Chemical structure is the floor's economy: it is cheaper, for bounded things carrying uncommitted boundary, to be joined than to be separate—and the same floor that makes joining cheaper makes the joint never collapse to one. That is why there are chemical structures at all.

8.1 Scope and limits

The account derives the qualitative architecture—existence of bonds, integer valence, closure rules, definite length and shape, reaction direction and equilibrium—from one functional and the inherited shell structure. It does not, and does not attempt to, compute bond energies, lengths, or angles numerically; those sit inside the constant κ and the convex profile $B(r)$, which we have characterised only by their qualitative properties (monotone in vacancy, convex in r, floored below). The honest division of labour is: the framework states the quantity chemistry minimises and reads off what must be true of its minimisers; electronic-structure theory computes the minimiser's coordinates. Three specific limits are worth marking. First, the covalent/ionic distinction is treated here only through the same vacancy-sharing arithmetic (NaCl and HCl differ in how unequally the shared cells are held, which we have not quantified). Second, multiple bonding (the double interfaces in CO_2, the delocalised interfaces in benzene) is handled at the level of cell-counting, not of the energetics that decide when a double interface beats two singles. Third, the geometry theorem gives the maximal-separation configuration but not the quantitative angle corrections from unequal regions, which we have only indicated. Each is a place where the qualitative derivation meets the quantitative theory it does not replace.

9 Conclusion

Taking the partition structure of the isolated atom as given, and adding only the contact floor $\beta > 0$—itself forced by the non-completability of the whole against which an atom is individuated—we have derived why there are chemical structures at all. A bond is the recognition that two bounded atoms can be individuated by one shared boundary more cheaply than by two separate ones (Theorem 3.4); it exists exactly when the join lowers total boundary thickness, which requires uncommitted valence—vacancy—to share. Valence is the vacancy count and the octet/duet rules are the per-atom thickness minimum (Theorem 4.1); bond length is the convex thickness minimiser, with the floor supplying the repulsive wall (Theorem 5.1); molecular shape is the maximally separated angular configuration that minimisation must take in three dimensions (Theorem 6.4); and reaction is the monotone, irreversible propagation of boundary toward a floor-limited minimum, with affinity the available reduction and equilibrium the halt (Theorem 7.2). These are not five rules but one minimisation seen five ways. The worked cases—H_2 and the inert noble gases, NaCl, HCl and LiF, H_2O, NH_3, CH_4, CO_2, and benzene—show the single principle fixing existence, stoichiometry, and shape together, with no parameter beyond the shells of the isolated atoms. The final statement is the simplest: a world of bounded atoms carrying uncommitted boundary is compelled to be a world of molecules, because for such atoms it is cheaper to share a boundary than to maintain two, and the floor that makes sharing cheaper also forbids the shared structure from ever collapsing to a point. Chemical structure is the economy of the floor, one level above the atom.

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